

1) Determine the multiplicative inverse module 26 for each of the following integers

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

$3^{-1} \text{ mod } 26$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1)$$

$$2 = 26 + 3(-8)$$

$$1 = 3 + 2(-1)26 + (8)3$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = 9$$

$5^{-1} \text{ mod } 26$

$$26 = 5(5) + 1$$

$$1 = 26 + 5(-5)$$

$$1 = 26 + 5(-5)$$

$$5^{-1} = 21$$

$7^{-1} \text{ mod } 26$

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$5 = 26 + 7(-3)$$

$$2 = 7 + 5(-1)$$

$$1 = 5 + 2(-2)$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} = 15$$

$$9^{-1} \text{ mod } 26$$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$(1) [9^{-1} = 3] + 8 = 1$$

$$(3) 8 + 0 - 26 + 8 = 1$$

$$11^{-1} \text{ mod } 26$$

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} = 19$$

$$15^{-1} \text{ mod } 26$$

$$26 = 15(1) + 11$$

$$11 = 26 + 15(-1)$$

$$15 = 11(1) + 4$$

$$4 = 15 + 11(-1)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$(3) [4(-1) + 26] + 3(-1) + 11(-1) + 26 = 1$$

$$6(0)P1 + (4+3(-1)) = 1 - 2S + (7)P1$$

$$15 + 11(-1) + (11 + 4(-2)) * (-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + (26 + 15(-1))(-2) = 1$$

$$15(7) + 26(-4) = 1$$

$$15 = 7 \quad (15 + 26) + 15$$

17-

$$26 = 17(1) + 9$$

$$9 = 26 + 17(-1)$$

$$17 = 9(1) + 8$$

$$8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26(-3) + 23(1) = 1$$

$$26(-4) + 23(2) = 1$$

$$26(-5) + 23(3) = 1$$

19-

$$26 = 19(1) + 7$$

$$7 = 26 + 19(-1)$$

$$19 = 7(2) + 5$$

$$5 = 19 + 7(-2)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + (7 + 5(-1))(-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + 7(-4) + 5(2) = 1$$

$$19 + (26 + 19(-1))(-4) + (19 + 7(-2))(2) = 1$$

$$19 + 26(-4) + 19(4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + (26 + 19(-1))(-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1 \quad (1-7)11+21$$

$$19+1=20 \quad (1-7)11+21$$

$$19(11) + 26(-8) = 1$$

$$19(21) + 26(-11) + 26(-11) + 19(11) = 1$$

$$26 = 21(1) + 5 \quad 5 = 26 + 21(-1)$$

$$21 = 5(4) + 1 \quad 1 = 21 + 5(-4)$$

$$21 + 5(-4) = 1$$

$$21 + (26 + 21(-1))(-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21 = 5 + (4)1 = 25$$

$$5 + (4)1 = 25$$

$$1 + (4)8 = 33$$

$$23 = 1$$

$$26 = 23(1) + 3 \quad 3 = 26 + 23(-1)$$

$$23 = 3(7) + 2 \quad 2 = 23 + 3(-7)$$

$$3 = 2(1) + 1 \quad 1 = 3 + 2(-1)$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + (23 + 3(-7))(-1) = 1$$

$$26 + 23(-1) + 23(-1) + (26 + 23(-1))(7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$17 + 23(-1) = 26$$

$$26 = 17 + 23(-1)$$

$$17 + 23(-1) = 26$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$1 = (5-1)26 - 1 = 25(-1) + (5-1)1 + 1$$

$$1 = (5-1)26 + (5-1)1 + (5-1)1 + 1$$

$$1 = 25(-1) + 25 + (5-1)1 + 1$$

$$1 = (5-1)26 + (5-1)1 + (5-1)1 + 1$$

$$1 = (5-1)26 + (5-1)1 + (5-1)1 + 1$$

$$1 = (5-1)26 + (5-1)1 + (5-1)1 + 1$$

- 2) Choose any string of your choice to encrypt using the affine cipher with key values multiplier 5 & additive 8

PLAINTEXT: ANSHUL

0 13 18 7 20 11

Formula: $C = (P \cdot \text{Multi} + \text{Add}) \bmod 26$

$$C = (P \times 5 + 8) \bmod 26$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 \quad T$$

$$N = (13 \times 5 + 8) \bmod 26 = 203 \bmod 26 = 21 \quad V$$

$$S = (18 \times 5 + 8) \bmod 26 = 278 \bmod 26 = 18 \quad S$$

$$H = (7 \times 5 + 8) \bmod 26 = 113 \bmod 26 = 17 \quad R$$

$$U = (20 \times 5 + 8) \bmod 26 = 308 \bmod 26 = 22 \quad W$$

$$L = (11 \times 5 + 8) \bmod 26 = 173 \bmod 26 = 11 \quad L$$

CIPHER TEXT: TVSRWL

- 3) Encrypt 'HI' using affine cipher with $A=11$, $b=5$

$$C = (P \times a + b) \bmod 26$$

$$H=7$$

$$C = (P \times 11 + 5) \bmod 26$$

$$I=8$$

$$C = H = (7 \times 11 + 5) \bmod 26 = (77 + 5) \bmod 26 = 82 \bmod 26 = 4 = E //$$

$$I = (8 \times 11 + 5) \bmod 26 = (88 + 5) \bmod 26 = 93 \bmod 26 = 15 = P //$$

Ciphertext: EP

- 4) The Ciphertext UCR was encrypted using the affine function $(9x+2) \bmod 26$ find the plaintext

$$U=20, C=2, R=17$$

$$C = (9P + 2) \bmod 26$$

$$C - b = 9P \bmod 26$$

$$C - b = 9P \bmod 26$$

9⁻¹

$$26 \equiv 9(2) + 8$$

$$9 \equiv 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$2S \text{ base } (b) 9^{-1} = 3 \cdot 10^{11} (9) = 1$$

$$U = 39(20-2) \bmod 26 = 39(18) \bmod 26 = 54 \bmod 26 = 2C$$

$$C = 39(2-2) \bmod 26 = 39(0) \bmod 26 = 0 = A$$

$$R = 39(17-2) \bmod 26 = 39(15) \bmod 26 = 45 \bmod 26 = 19 = T$$

$$V = 1S = (2S \text{ base}) 80S = 2S \text{ base } (8 + 21 \times 81) = 14$$

$$2 \quad 81 = (2S \text{ Plaintexts } CA + \text{base } (2 + 21 \times 81) = 2$$

$$8 \quad 81 = (2S \text{ base}) 81 = 2S \text{ base } (2 + 21 \times 81) = 14$$

5) Decrypt the ciphertext using the affine cipher
KDXXTUNDEXYZ using the key 13(Mult) & 4

$$a=3, b=4$$

$$P = a^{-1}(C - b) \bmod 26$$

2nd 1=A K D Y X T U W P D E X Z Y

10 3 24 23 9 20 22 3 4 23 25 24

$$a=3 \quad 2S \text{ base } (2 + 21 \times 9) = 1$$

$$b=4 \quad 2S \text{ base } (2 + 21 \times 9) = 1$$

$$2S \text{ base } 88 = 2S \text{ base } (2 + 21 \times 26) = 3(2) + 2(2 + 21 \times 2) = 26 + 3(-2)$$

$$3 = 2(1) + 1$$

$$2S \text{ base } 8P = 2S \text{ base } (2 + 21 \times 2) = 3 + 2(-1) = 1 \text{ base } (2 + 21 \times 2) = 1$$

$$1 = 3(1) = 3 + (26 + 3(-2))(-1) = 1$$

$$3 + 26(-1) + 3(2) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} + 9 = 9^{-1} \bmod 26$$

$$2S \text{ base } (2 + 21 \times 9) = 1$$

$$K = 9(10-4) \bmod 26 = 9(6) \bmod 26 = 2 = C$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$Y = 9(24-4) \bmod 26 = 9(20) \bmod 26 = 24 = Y$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$J = 9(9-4) \bmod 26 = 9(5) \bmod 26 = 19 = T$$

$$U = 9(20-4) \bmod 26 = 9(16) \bmod 26 = 14 = O$$

$$W = 9(22-4) \bmod 26 = 9(18) \bmod 26 = 6 = G$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 = R$$

$$F = 9(4-4) \bmod 26 = 9(0) \bmod 26 = 0 = A$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 = P$$

$$Z = 9(25-4) \bmod 26 = 9(21) \bmod 26 = 7 = H$$

$$Y = 9(44-4) \bmod 26 = 9(40) \bmod 26 = 24 = Y$$

Plaintext: CRYPTOGRAPHY

- 6) Use $K_1 = 13$ (Multi), $K_2 = 4$ (Even though it is not a valid key term to encrypt). Encrypt the string 'input' & 'alter' above key & find what is write.

$$C = (P \times \text{Multi} + \text{Add}) \bmod 26$$

$$C = (P \times a + b) \bmod 26$$

$$a = 13 \quad b = 4$$

13 is not co-prime of 26

So we cannot find inverse of 13 mod 26

Therefore it is not a valid key for affine cipher

For 'input'

$$I = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

ERRER

For 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = R$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R$$

ERRER

Both the plaintext & ciphertext will produce same text.

The encryption is not 1:1 & decryption is not possible as 13 is not a valid key.

2) Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 9 \end{bmatrix} \bmod 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 9 \end{bmatrix} \bmod 26$$

$$\det(K) = 3 \times 9 - 2 \times 5 = 27 - 10 = 17$$

17^{-1}

$$26 = 17(2) + 4 \quad 4 = 26 + 17(-2)$$

$$17 = 4(2) + 3 \quad 3 = 17 + 4(-2)$$

$$4 = 3(1) + 1 \quad 1 = 4 - 3(1)$$

$$4 + 3(1) = 1 \quad 1 = 4 + 3(1)$$

$$26 + 17(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 17(-2) + 11(-1) + 4(2) = 1$$

$$26 + 17(-3) + 26(2) + 11(-4) + 4(1) = 1$$

$$26(3) + 17(-7) = 1 \quad (13 \times 3 + 17 \times -7) = 1$$

$$26(7) + 17(-9) = 1 \quad (17 \times 7 + 26 \times -9) = 1$$

$$26(18) + 17(-19) = 1 \quad (17 \times 18 + 26 \times -19) = 1$$

Adjoint of the matrix

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26 = \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

3) Find the inverse of the following matrix whose entries are considered modulo 26: $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \pmod{26}$$

$$\det(K) = 11 \times 3 - 13 \times 2 = 33 - 26 = 7$$

$$7^{-1} \quad 26 = 7(3) + 5 \quad 5 = 26 + 7(-3)$$

$$7 = 5(1) + 2 \quad 2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$3 + 2(-2) = 1$$

$$26 + 7(-3) + 2(7 + 5(-1))(-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + (26 + 7(-3))(2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-1) = 1$$

$$26 - 11 = 15$$

Adjoint is

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix} //$$

9) Is the key valid?

$$K = \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

For a key to be valid - 1) determinant should be co-prime & should not be equal to 0

$$\det(K) = 2(21) - 14(13) = 42 - 182 = -140$$

$$-140 \bmod 26 = 16$$

$$\det K = 16 \bmod 26$$

16 is not co-prime to 26

∴ It will not have an inverse module 26

∴ The key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid.

10) Encrypt using 2x2 Hill cipher for the plaintext 'SOLVED' using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 = 3 \text{ which is co-prime}$$

$$S O | L V | E D \quad C = K \times P$$

$$P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 = 10 = K$$

$$280 \bmod 26 = 20 = U$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 8 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 162 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26$$

$$201 \bmod 26 = 19 = T$$

$$308 \bmod 26 = 22 = W$$

$$\begin{matrix} E \\ D \end{matrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} \bmod 26 = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26$$

$$36 \bmod 26 = 10 = K$$

$$61 \bmod 26 = 9 = J$$

SOLVED $\rightarrow$ KUTWKJ

$$001 - (81 - 54) = (21)1 - (15)5 = (6)6$$

$$21 - 25 \bmod 26 = 1$$

$$25 \bmod 26 = 25$$

is a not co-prime to 26

It will not have an inverse modulo 26

$$\begin{bmatrix} 1 & 5 \\ 13 & 21 \end{bmatrix} \text{ not valid}$$

(1) Encrypted word XYS will cipher for the plaintext

$$\begin{bmatrix} 2 & 2 \\ 8 & 8 \end{bmatrix}$$

$$2 \times 1 - 1 \times 8 = -3 = 3 \text{ which is co-prime}$$

$$20/24 = 5 \quad 2 \times 5 = 10$$

$$\begin{bmatrix} 1 & 1 \\ 13 & 21 \end{bmatrix} \cdot \begin{bmatrix} 2 & 2 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} 14 & 10 \\ 28 & 18 \end{bmatrix} \bmod 26 = \begin{bmatrix} 14 & 10 \\ 2 & 18 \end{bmatrix}$$

$$14 \bmod 26 = 14$$

$$10 \bmod 26 = 10$$

$$\begin{bmatrix} 14 & 10 \\ 2 & 18 \end{bmatrix} \cdot \begin{bmatrix} 1 & 5 \\ 13 & 21 \end{bmatrix} = \begin{bmatrix} 14+50 & 294+210 \\ 26+182 & 42+378 \end{bmatrix} \bmod 26 = \begin{bmatrix} 18 & 10 \\ 10 & 18 \end{bmatrix}$$

$$18 \bmod 26 = 18$$

$$10 \bmod 26 = 10$$

$$\begin{bmatrix} 18 & 10 \\ 10 & 18 \end{bmatrix} \cdot \begin{bmatrix} 2 & 2 \\ 8 & 8 \end{bmatrix} = \begin{bmatrix} 36+80 & 108+162 \\ 20+144 & 80+144 \end{bmatrix} \bmod 26 = \begin{bmatrix} 16 & 18 \\ 18 & 16 \end{bmatrix}$$